Pneumonia Detection using Chest X-Ray Images with Custom CNN

1. Abstract

Pneumonia is a severe respiratory infection that needs to be diagnosed early and accurately to decrease the mortality, particularly in children and elderly patients. Chest X-ray imaging is one of the most commonly used diagnostic tools, but manual interpretation is time-consuming and prone to human error. This project showcases an end-to-end deep learning pipeline for automatic pneumonia detection from chest X-ray images based on a custom-made Convolutional Neural Network (CNN). The whole workflow involves data loading, data preprocessing, data augmentation, model architecture design, model training, model validation, model testing, and model performance evaluation. No pretrained model has been used, the CNN was trained from scratch. Experimental results show that the proposed model can be used to effectively separate the Normal and Pneumonia cases, showing a good performance with strong interpretability and reproducibility.

1. . Problem Identification

Who is affected?

Pneumonia affects millions of people around the world every year, and its risk is seen to be at its highest among young children, the elderly, and people with damaged immune systems. In many areas, particularly low-resource areas, access to experienced radiologists is low resulting in delayed or incorrect diagnosis.

Why is this a problem?

Delayed or improper diagnosis of pneumonia can lead to severe complications, prolonged hospitalization or even death. Chest X-rays are common diagnostic tools, however, they are subjective and take a long time to interpret. Automating the initial screening process can help assist clinicians and reduce the workload and increase efficiency in the diagnosis process.

Specific unmet need in health care

There is a need for an inexpensive, rapid and dependable decision support system to help healthcare professionals diagnose pneumonia from chest x-ray images especially in settings where expert radiologists are not always available

2. Dataset Justification

The data set used in this project is labeled chest x-ray images divided into the Normal and Pneumonia class. Chest X-rays are the standard imaging modality for the diagnosis of pneumonia therefore this data has a lot of relevance in the problem domain.

Data structure in the dataset is for supervised learning and it is easy to split the data into train, validation, and testing data. In fact, it is fitting for this task because:

- It reflects real-world clinical imaging data
- Labels are clinically meaningful
- Image-based learning is congruent with diagnostic practice

The dataset facilitates testing for generalization of models with reproducibility.

3. Methodology

Data Preprocessing Pipeline

- Image resizing to 224x224 pixels
- Normalization of pixel intensities
- Data augmentation on data (rotation, flipping, cropping, brightness)
- No augmentation applied to validation & test sets

Model Architecture

A Convolutional Neural Network (CNN) model was created and trained from scratch. The architecture includes:

- Multiple convolutional layers with increasing filter depth
- Batch Normalization & ReLU Activation
- Max pooling layers
- Fully connected layers using dropout

This is a compromise between the model capacity and computational efficiency. (Architecture diagrams may be included in the final submission.)

Training Process

Optimizer: Adam

Learning rate: 0.001

Loss function: Binary Cross-Entropy _ logits

Epochs: Up to 25

Class weighting to deal with imbalance

Validation Strategy

A different validation set was used to monitor the performance of the model during training and avoid overfitting. Final assessment was done on a test set held out.

4. Usage & Adaptation of Pretrained Models

Rationale

No pretrained models were used in this project. As per hackathon rules and to make the model fair and original, all parameters of the model were randomly initialized and the model was trained from scratch.

Modifications

Not applicable because no pretrained architecture was used.

Training Strategy

The whole network was trained end-to-end from scratch and with a single learning rate.

Risk & Bias Discussion

Training from scratch prevents any bias from large-scale natural image datasets. However, there is still a risk for biases to occur from:

- Dataset imbalance
- Limited diversity of imaging conditions

Care was taken to address such issues with data augmentation and class-weighted loss.

5. Results

Metrics

The model was tested using:

- Accuracy
- Precision
- Recall
- F1-score

Visualizations

- Training loss curves
- Confusion matrix

Error Analysis

Misclassifications were mainly found in borderline cases where visual patterns between Normal and Pneumonia images overlap.

Limitations

- Binary classification only
- Performance that depends on quality of image
- Limited dataset diversity

6. Real-world Application

Proposed Scenario for Deploying

The system may be implemented as a clinical decision support tool that is integrated into the radiology systems of the hospital.

Potential Users

- Radiologists
- General physicians
- Hospitals in resource constrained settings

Workflow Integration

Chest X-rays are uploaded and the system returns a value that is a probability of pneumonia to help the clinicians.

Risks & Limitations

- Not a substitute for medical experts
- Needs clinical validation before deployment

7. Marketing & Impact Strategy

Adoption

Hospitals, diagnostic centers and public healthcare providers could use the system.

Practical Benefits

- Faster diagnosis
- Reduced clinician workload
- Improved accessibility

Cost & Accessibility

The model can be executed on an off-the-shelf hardware, which makes it applicable for low cost deployment.

8. Future Improvements

Model Enhancements

- Multi-class classification
- Methods of explainability (Grad-CAM)

Additional Data Needs

- Larger and more diverse data sets

Clinical Translation Pathways

- Prospective clinical studies
- Regulatory approvals

9. Conclusion

This project provides a full and repeatable machine learning pipeline for pneumonia detection in chest X-ray images. And, by training a custom CNN from scratch, the system follows hackathon constraints and solves an important healthcare need. The approach shows great promise as a decision support tool that has a future for clinical translation.

This project shows the feasibility of creating a full medical image classification pipeline using a custom CNN model that is trained from scratch. The results highlight the potential of deep learning in aiding the diagnosis of pneumonia alongside the importance of reproducibility, ethical considerations, and scientific rigor in healthcare applications of AI.